

TrES-1b

R-1558

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_220604 · created 2026-08-01T20:57:01+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459734.7891 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.049
Transit depth (Rp/Rs)^2	3.37 +/- 1.81 [%]
Orbital Inclination (inc)	87.24 +/- 1.37
Airmass coefficient 1 (a1)	1.128 +/- 0.068
Airmass coefficient 2 (a2)	-0.091 +/- 0.044
Scatter in the residuals of the lightcurve fit is	6.13 %
Best Comparison Star	#2 - [361, 426]
Optimal Aperture	5.71
Optimal Annulus	14.7
Transit Duration (day)	0.095 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.049 vs 0.1358 (deviation 0.74σ)
Duration fit vs published	0.095 d vs 0.1042 d (deviation 0.49σ)
Mid-time O–C	1.5 min
Depth SNR	2.8
Residual scatter	6.13 %

Light curve

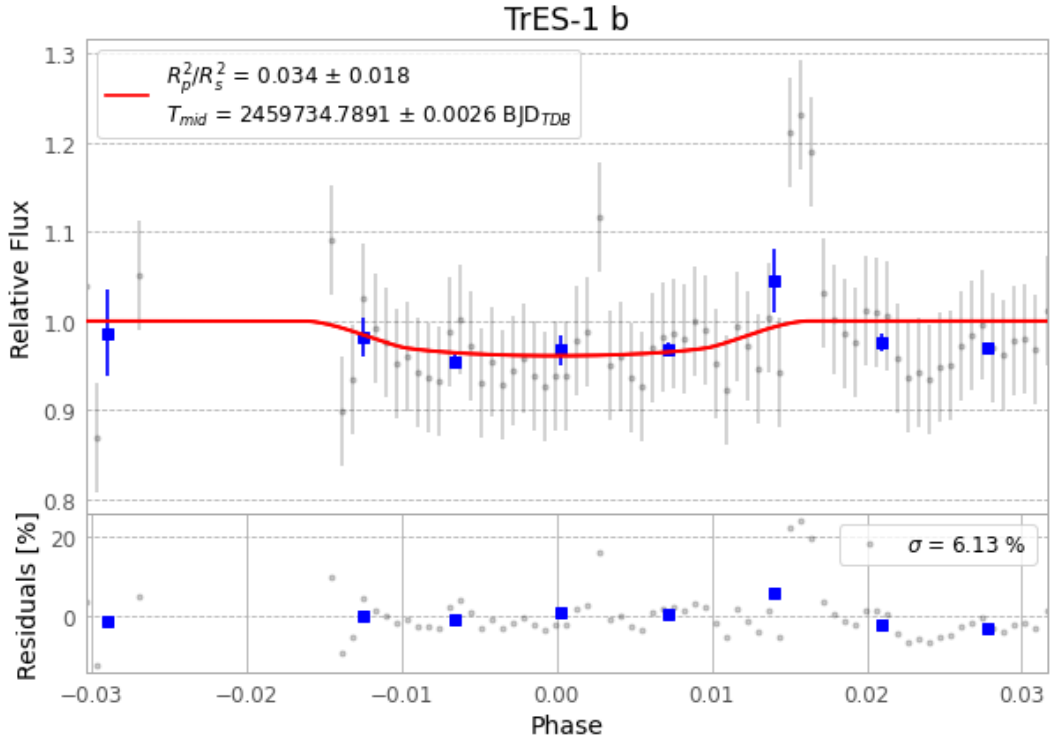


Figure 1 — FinalLightCurve_TrES-1 b_2022-06-03.png (SHA-256 1a97d9ebe7d4592d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [634, 248], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b9768a411abad6f9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

91 FITS frames (60 MB), TRES-1220604043912.FITS → TRES-1220604090913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-220604.log (d99b7f3665de...), FinalLightCurve_TrES-1 b_2022-06-03.png (1a97d9ebe7d4...), FinalLightCurve_TrES-1 b_2022-06-03.csv (df4536b26ec4...)

Compute resources

Wall time 200.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 20:57Z.